

GD2 CAR-T toxicity

Clinical and preclinical safety, intracerebroventricular delivery, and transient organ-function change relevant to concomitant therapy

Literature synthesis of 852 curated records (2003–2026)

GD2 CAR-T toxicity, with emphasis on intracerebroventricular delivery

Synthesis of **852** curated records (2003–2026, 502 with open-access full text) on GD2 CAR-T safety. Domain detail is in [notes/](#); this report states the answers, the evidence behind them, and what is not known.

1. Bottom line

1. **On-target/off-tumour toxicity has not materialised clinically.** Across every published GD2 CAR-T/CAR-NKT cohort, no patient has developed the painful neuropathy that anti-GD2 antibodies cause in most children, and no radiographic or clinical evidence of normal-CNS injury has been attributed to the CAR. Autopsy in one DIPG patient found CAR transcript and lymphocytic infiltrate confined to tumour, with tumour GD2 $\gg$ normal brain GD2 (PMID 35130560). The mechanism is antigen-density discrimination — and it is fragile: raising scFv affinity (14G2a E101K) caused **fatal encephalitis in low-GD2 cerebellar and basal brain regions** in mice (PMID 29180536).
2. **The dominant clinical toxicity is inflammation, and its location depends on the route.** IV infusion produces CRS (dose-limiting at $3 \times 10^6/\text{kg}$ in DMG: three grade 4 events) plus ICANS in $\sim 45\%$ of patients. ICV infusion produces almost none of either but near-universal **TIAN** on first exposure.
3. **ICV administration without lymphodepletion is markedly less toxic systemically.** In the only internal comparison (NCT04196413 arm A; 11 IV vs 62 ICV infusions): **no DLT in 62 ICV infusions**, CRS in 33.9% of ICV infusions (mostly grade 1) vs 100% of IV infusions, and **zero ICANS after ICV** despite higher CSF cytokines (PMID 39537919). The 253-dose ICV B7-H3 experience agrees: commonest AEs headache/nausea/fatigue/fever, one grade 3 hydrocephalus, no ICANS, one DLT (intratumoural haemorrhage) (PMID 39775044).
4. **The ICV-specific hazards are mechanical and hardware-related, not biochemical:** raised ICP (documented 22 and 34 mmHg episodes reversing within minutes of CSF drainage), obstructive and communicating hydrocephalus, reservoir/catheter dependence, and intratumoural haemorrhage of ambiguous attribution.
5. **Transient organ-function change is a function of systemic inflammation, and is therefore an IV problem.** No GD2 CAR-T trial reports attributed hepatic, renal or electrolyte toxicity; one trial explicitly documents normal liver enzymes, bilirubin and electrolytes acutely and long-term (PMID 40420236). In general CAR-T populations, hepatic dysfunction accompanies **64% of grade 3–4 CRS** and grade 3–4 transaminase elevation occurs in **31–64%** of severe CRS, while AKI occurs in **$\sim 18.6\%$** pooled and **resolves in $\sim 79\%$ within a month**.
6. **Cytopenias in GD2 CAR-T trials are lymphodepletion, not CAR, toxicity** — reversible in a median 4 days in one series, and occurring before cell infusion in the CAR-NKT trial. ICV re-dosing omits lymphodepletion and produced essentially no cytopenia (one grade 1 lymphocytopenia in 21 patients over 253 doses).
7. **Drug-interaction risk is real but indirect and unmeasured in this therapy.** CRS-grade inflammation suppresses CYP3A4/2C19/1A2 (midazolam clearance -65% as

CRP rose 10→300 mg/L; PBPK simvastatin AUC ×2.4 at IL-6 100 pg/mL), and IL-6 blockade reverses it (sarilumab cut simvastatin AUC by **45%**). The best transient-cytokine analogue predicts **~28% CYP3A4 suppression for about a week**. Nobody has measured a DDI in a CAR-T patient. The therapy-specific interactions that are documented are different in kind: renal function governs fludarabine exposure during conditioning, and corticosteroids blunt the CAR itself.

2. IV versus ICV, quantified

Same product, same trial, same patients (PMID 39537919; first four patients in PMID 35130560):

	IV (11 infusions)	ICV (62 infusions)
Lymphodepletion	Cy/Flu required	none
Dose	1×10 ⁶ /kg (MTD) or 3×10 ⁶ /kg	10–30×10 ⁶ flat
CRS	100% of infusions; grade 4 in 3/8 at DL2 → DLT	33.9%; 16 grade 1, 4 grade 2, 1 grade 3 (urosepsis)
ICANS	5/11 patients (grade 3 in 1)	0
TIAN	91–100% of patients; grade 3–4 in 5/11	71% of infusions; grade 3–4 in 8/62
Cytopenias	grade 3–4 expected (Cy/Flu)	not reported
Hepatic/renal/electrolyte AEs	none attributed (CRS mostly ≤grade 2 at MTD)	none attributed
Haemorrhage	1 grade 5 intratumoural (vascular anomaly)	—
DLTs	yes (CRS)	none
Blood cytokines	higher	lower
CSF cytokines	lower	higher

The dissociation — higher CSF cytokines yet no ICANS — is the most informative safety observation in the collection: ICANS tracks the blood/endothelial compartment, and moving the inflammatory response into the CSF does not reproduce it. It also explains why systemic organ-function change tracks the IV route.

3. Toxicity domains and evidence grade

Domain	GD2-specific evidence	Strength	Notes
CRS	6 clinical cohorts; near-universal after IV; grade 4 at DL2 in DMG; mild in 95% of GD2-CART01	Direct, consistent	Dose- and route-driven
TIAN	Defined in the GD2 DMG trial; 100% first ICV exposure; all reversible	Direct	Grading not prospectively validated
ICANS	5/11 in DMG IV; grade 3 in 4/54 on GD2-CART01 (rimiducid-reversed); 0/62 ICV	Direct	Distinguishable from TIAN only when tumour is focal
On-target/off-tumour	Absent in all cohorts; fatal in high-affinity preclinical model		Affinity/architecture dependent

Domain	GD2-specific evidence	Strength	Notes
		Direct-negative + preclinical hazard	
Haematologic	Near-universal grade 3–4 with Cy/Flu; pooled anaemia 97%, neutropenia ≥ 3 93%	Direct but confounded	Conditioning, not CAR
Hepatic	No attributed events; explicit normal labs in one trial	Indirect [CART-general]	Expect grade 1–2 ALT/AST if CRS ≥ 3
Renal/electrolyte	No attributed events; sodium moved deliberately for TIAN	Indirect [CART-general]	AKI ~18.6% pooled after systemic CAR-T
Coagulopathy	Not reported in GD2 cohorts	Indirect	Matters before neurosurgical procedures
Cardiopulmonary	Grade 4 CRS: pressors, pulmonary oedema, intubation; capillary leak 4/10 in one trial	Direct	CRS-mediated
IEC-HS/HLH	None reported; high ferritin without HLH	Indirect ; ~3.5% after CAR-T generally	The scenario where hepatic dysfunction becomes clinical
Infection	Fungal and bacterial cases; urosepsis confounded 2 CRS events	Direct	Stacked immunosuppression
Drug interactions	None measured	Mechanistic only	See §5
Device/hardware	Reservoir required; 1 grade 3 hydrocephalus in ICV B7-H3; haemorrhage DLT	Direct, ICV-specific	Under-reported as a category
Myeloid/microglial amplification	CSF scRNA-seq myeloid states differ by route; grade ≥ 2 TIAN correlates with CSF MCP-1/CCL2, TNF- α , IL-10	Direct but correlative	Mechanism untested; see notes/08

3a. The myeloid axis behind TIAN

TIAN correlates with myeloid, not T-cell, readouts: grade ≥ 2 TIAN tracks CSF **MCP-1/CCL2**, TNF- α and IL-10, and grade ≥ 2 CRS tracks plasma MCP-1/CCL2. CSF single-cell RNA-seq shows route-dependent myeloid states — an interferon-responsive population at peak inflammation after ICV, versus phagocytic, lipid-metabolic, immune-suppressive DAM/MDSC-like states after IV and at late timepoints. Autopsy in the first DMG patient showed dense Iba1⁺/CD163⁺ myeloid infiltration **within tumour** and resting, non-reactive microglia in unaffected cortex. Microglia are therefore best read as **amplifiers** downstream of tumour-restricted CAR engagement, not as GD2 targets; the causal claim remains untested. Detail in notes/08.

3b. Failure modes (the other axis)

Efficacy is limited without GD2-negative escape being the reason: pooled neuroblastoma CR is ~40% (PMID 41530803), GD2-CART01 reaches 66% ORR and 42.7% 5-year OS (77% / 68% in low-burden patients), while DMG responses are transient. Dominant failure modes are **pre-infusion attrition** (waitlist and manufacturing-window progression; worse outcomes with late apheresis and ≥ 3 prior lines), **exhaustion and short functional persistence** (tonic-signalling-prone 14g2a construct; plain GD2.CART benefit ≤ 3 weeks

versus longer PFS with C7R; BTG1-driven hyporesponsiveness in CAR-NKT), **compartmental delivery failure** (a temporal-lobe lesion with high uniform GD2 did not respond while ICV-treated spinal disease regressed), **MDSC/G-CSF-driven microenvironmental suppression**, and **iatrogenic** loss of activity (steroids, dasatinib, iCasp9 ablation, DL1 as MTD, no lymphodepletion with ICV). Detail in [notes/09](#).

4. Transient organ-function changes that could affect other therapies

Magnitude, timing and reversibility are tabulated in [notes/05](#). The decision-relevant summary:

- **Liver:** laboratory transaminase elevation is common and under-reported as an AE (58% laboratory vs 8% reported in one 148-patient trial); grade ≥ 3 is uncommon (0–8%) unless CRS is grade 3–4, where hepatic dysfunction reaches 64%; AST>ALT pattern suggests hypoperfusion; bilirubin usually spared outside IEC-HS. Resolves with CRS.
- **Kidney:** AKI ~18.6% pooled, onset day +1 to +28 (mode day +7), mostly mild, **79% recovered within a month**; predictors are CRS, ICANS grade ≥ 3 , pre-existing CKD; associated with worse survival even when reversible.
- **Electrolytes/fluid:** hyponatraemia from resuscitation/SIADH; **iatrogenic sodium shifts from hypertonic saline given for TIAN**; hypophosphataemia and hypokalaemia with inflammation; tumour lysis with high burden.
- **Marrow and haemostasis:** near-universal transient cytopenias with Cy/Flu, prolonged in a minority (ICAHT, predictable by CAR-HEMATOTOX); CRS coagulopathy common with low absolute bleeding rates but relevant to reservoir/EVD procedures during the inflammatory window.
- **Consequences for concomitant therapy:** nephrotoxins and contrast should be avoided in the CRS window and renally cleared drugs dosed on the current creatinine; myelosuppressive bridging or concurrent chemotherapy compounds ICAHT and should be sequenced rather than overlapped; neurosurgical procedures should be planned around coagulation parameters; steroid-sparing is preferred both for efficacy and to avoid steroid-mediated interactions.
- **After ICV dosing without lymphodepletion, none of this applies to a measurable degree** — the main systemic constraint on co-administered therapy is removed.

5. Drug-interaction assessment (explicitly graded)

What is established [mechanistic-PK]: acute inflammation suppresses CYP-mediated clearance in a CRP/IL-6-dependent way (midazolam CL –65% over CRP 10–300 mg/L; postoperative CYP3A4 –61%, CYP1A2 –53%, CYP2C19 –58%; ciclosporin C_{ss} $\times 3.6$ at IL-6 peak; darunavir CL/F halved in COVID-19 with IL-6 the sole covariate). IL-6 blockade reverses it (sarilumab: simvastatin AUC –45%, C_{max} –46%; sirukumab: midazolam AUC –30–35%). For a **transient** cytokine spike the modelled effect is **<2-fold, ~28% CYP3A4 suppression, lasting ~1 week** (blinatumomab analogue).

What is not established: any measured interaction in a CAR-T, GD2 CAR-T, or ICV CAR-T patient. No CYP-probe study exists in CRS. The magnitude in CAR-T CRS is likely to exceed the blinatumomab model (higher, longer cytokine elevation) but this is inference.

Therefore: treat CRS as a transient, bidirectional perturbation of CYP3A4-dominant clearance — levels up during CRS, down after IL-6 blockade — and manage it with therapeutic drug monitoring for narrow-index substrates (azoles, calcineurin/mTOR inhibitors, anticonvulsants, anticoagulants) rather than with pre-emptive dose changes. Do not claim a GD2-specific interaction.

Therapy-specific interactions that are documented, and matter more in practice:

- **Renal function → fludarabine exposure** (eGFR a significant covariate on clearance in the first CAR-T-specific population-PK model, PMID 41471106): a transient creatinine rise before conditioning changes lymphodepletion intensity, which changes both efficacy and toxicity.
- **Corticosteroids → CAR activity:** a pharmacodynamic interaction the trials manage explicitly (steroid-free at enrolment; response scoring deferred ≥ 7 days after steroids).
- **Anti-cytokine agents → infection and marrow recovery:** anakinra, tocilizumab, siltuximab, and, for IEC-HS, etoposide/ruxolitinib/emapalumab stack with lymphodepletion.
- **Hypertonic saline** for TIAN as a deliberate electrolyte intervention interacting with volume- and sodium-sensitive drugs.

6. What would change these conclusions

1. Serial laboratory and cytokine data published for an **ICV-only** cohort (arms B/C of NCT04196413) — would convert "no reported organ toxicity" into a measured claim.
2. A **CYP-probe or population-PK study around CRS and tocilizumab** in CAR-T recipients.
3. Prospective **TIAN grading** with the type 1/type 2 distinction operationalised, since BrainChild-03 could not classify most events retrospectively (PMID 41798119).
4. A larger ICV denominator for **intratumoural haemorrhage** attribution.
5. A circulating **neuro-injury biomarker** (e.g. NFL) in GD2 CAR-T patients, to test the on-target/off-tumour null result with something more sensitive than examination.
6. Any clinical use of a **higher-affinity or armoured GD2 CAR** — the preclinical hazard is specific to affinity/architecture and the current safety record should not be assumed to transfer.

7. Practical distillation

- The ICV route is the main safety lever: it converts a systemic-inflammatory toxicity problem into an intracranial-pressure and hardware problem, and removes the lymphodepletion-driven organ and marrow toxicity that otherwise constrains concomitant treatment.
- Every ICV programme needs a reservoir, ICP measurement, an osmotic-therapy and CSF-drainage pathway, and staff who will access the reservoir within minutes — not a pharmacological algorithm alone.
- Grade CRS, ICANS and TIAN separately at every infusion; fever after ICV dosing is expected.

- Assume liver enzymes and creatinine will move transiently whenever CRS reaches grade 3, monitor them daily in the acute window, and use those values (not admission values) for co-medication dosing.
- Handle drug interactions as monitoring problems in a moving inflammatory window, and say plainly that the GD2-specific interaction data do not exist.

00 — Overview: what "GD2 CAR-T toxicity" means, and why the route is the question

The three toxicity engines

GD2 CAR-T toxicity is not one thing. Three mechanisms produce overlapping syndromes and are treated differently:

1. **Systemic cytokine release** — CAR T cells activate, myeloid cells amplify (IL-1, IL-6, MCP-1), and the patient develops fever, hypotension, hypoxia, endothelial activation, transient organ dysfunction. Downstream of this sit CRS, IEC-HS/HLH, coagulopathy, hepatic and renal changes, and most of the drug-interaction risk. Dose- and route-dependent.
2. **Inflammation at the tumour site** — in CNS tumours the antitumour response happens inside a closed box. Oedema, mass effect, CSF-outflow obstruction and local electrophysiologic dysfunction produce **TIAN** (tumour inflammation-associated neurotoxicity), which is distinct from ICANS: symptoms map to tumour neuroanatomy, and the treatment is CSF drainage / hypertonic saline / anti-cytokine therapy rather than steroids alone (Majzner 2022, PMID 35130560).
3. **On-target / off-tumour engagement of normal GD2** — GD2 is expressed on peripheral nerve, some CNS neurons and melanocytes. With antibodies this reliably produces neuropathic pain; with CAR T cells it has been essentially absent clinically at standard affinity, but is fatal in preclinical models when scFv affinity is raised (Richman 2018, PMID 29180536).

A fourth, mostly non-CAR effect matters practically: **lymphodepletion**. Cyclophosphamide/fludarabine causes the grade 3–4 cytopenias reported in most GD2 CAR-T trials, and ICV re-dosing is typically given without lymphodepletion — which is a large part of why ICV looks safer.

Why the route dominates the answer

For H3K27M+ DMG the same product has now been given both ways in the same trial (NCT04196413, arm A), so the comparison is internal rather than cross-trial (Monje 2025, PMID 39537919; 11 patients, 11 IV and 62 ICV infusions):

Toxicity	IV (DL1 $1 \times 10^6/\text{kg}$, DL2 $3 \times 10^6/\text{kg}$)	ICV ($10\text{--}30 \times 10^6$ flat, no lymphodepletion)
CRS	100% of infusions; grade 4 in 3/8 at DL2 (dose-limiting)	33.9% of infusions; 16/21 grade 1, one grade 3 (in urosepsis)
ICANS	5/11 patients (1 grade 3)	0 of 62 infusions
TIAN	100% DL1, 87.5% DL2; grade 3–4 in 5/11	71% of infusions; grade 3–4 in 8/62
Dose-limiting toxicity	Yes — CRS defined DL1 as MTD	None across 62 infusions

The route changes which compartment inflames. CSF cytokines are **higher** after ICV; blood cytokines and CRS are higher after IV. Systemic organ-function toxicity and ICANS track the blood compartment; TIAN and intracranial pressure track the CSF/tumour compartment.

Evidence taxonomy used in these notes

Claims are labelled by source strength, because the temptation in this field is to import CD19-lymphoma numbers into a 30-patient pediatric CNS experience:

- **[GD2-clinical]** — from a GD2 CAR-T/CAR-NKT trial. Small n, phase 1, mostly pediatric.
- **[ICV-clinical]** — from an ICV/locoregional CAR-T trial of any target (GD2, B7-H3, IL13R α 2, EGFR806, bivalent EGFR/IL13R α 2). Route-specific but often not GD2-specific.
- **[CART-general]** — from CD19/BCMA/CD22 experience. Large n, adult-dominant, systemic route; transfers for mechanism and for lymphodepletion-related toxicity, over-predicts for ICV.
- **[GD2-antibody]** — dinutuximab/naxitamab/hu14.18/3F8. Defines the on-target-off-tumour ceiling for the antigen but not for the modality.
- **[preclinical]** — mouse/xenograft. Predicted the brainstem-oedema/hydrocephalus problem and the affinity/CNS-toxicity relationship, so it is not decorative here.
- **[mechanistic-PK]** — inflammation/cytokine effects on drug-metabolising enzymes, mostly from rheumatoid arthritis, surgery, sepsis, COVID-19 and PBPK modelling. **No GD2 CAR-T study has measured a drug-drug interaction**; this literature supports plausibility and monitoring, not a quantified GD2-specific interaction.

What the collection says in one paragraph

At standard affinity and clinically used doses, GD2 CAR T cells have not produced the on-target off-tumour neurotoxicity that preclinical high-affinity constructs predicted; the dominant clinical toxicities are CRS (IV, dose-limiting), TIAN (both routes, near-universal in DIPG but reversible with a defined algorithm), lymphodepletion-driven cytopenias, and — in neuroblastoma — grade 3 ICANS in a minority, aborted by iCasp9 activation. ICV administration without lymphodepletion largely removes systemic CRS, ICANS and measurable hepatic/renal/haematologic derangement, and concentrates risk into intracranial pressure, hydrocephalus, device-related events and intratumoural haemorrhage. Transient organ-function change — and therefore concomitant-drug risk — is essentially a function of systemic inflammation: it is real after IV infusion (hepatic dysfunction in up to 64% of grade 3–4 CRS, AKI ~18% pooled after systemic CAR-T, resolving in ~79% within a month) and near-absent after ICV dosing.

01 — GD2 CAR-T clinical toxicity, trial by trial

All datasets below are phase 1 or phase 1/2, mostly pediatric, single- or few-centre. Toxicity is reported per patient unless stated per infusion.

H3K27M+ DMG / DIPG — GD2-CAR, 4-1BB, iCasp9 (Stanford, NCT04196413)

First four patients, IV at DL1 with a second dose ICV (Majzner 2022, PMID 35130560) [GD2-clinical, ICV-clinical]

- CRS after IV: grade 1 (patient 1), grade 2 (patient 2, fluid-responsive hypotension), grade 1 (patient 3), grade 3 in the eIND spinal DMG patient (vasopressor-requiring, day +6, tocilizumab + 3 days steroids).
- TIAN after IV in all four: worsening of existing deficits mapping to tumour location — cranial nerve palsies, trismus with T2/FLAIR signal in the trigeminal nucleus, transient hearing loss (2 days), increased ataxia.
- Patient 1: day +9 acute deterioration (hemiplegia, extensor posturing) with **ICP 22 mmHg**; 10 mL CSF drained via Ommaya → return to baseline "within minutes"; anakinra started; MRI showed increased pontine oedema.
- After ICV re-dosing (30×10^6 , no lymphodepletion): fever 40 °C within 24 h, then at ~48 h somnolence and new third-nerve palsy with **ICP 34 mmHg** and obstructive hydrocephalus from fourth-ventricle compression; continuous CSF drainage + hypertonic saline + anakinra + steroids; hydrocephalus resolved in 2 days, steroids stopped by day +6.
- The 5-year-old's ICV dose (12.9×10^6) caused **only mild headache**, no anti-cytokine agents, no steroids — i.e. ICV toxicity is not obligatory.
- Spinal DMG patient's first-ever ICV infusion (50×10^6 after intensified lymphodepletion): persistent fever, grade 3 encephalopathy graded as **grade 4 ICANS**, EEG with diffuse slowing and triphasic waves, no MRI change in normal brain; ICANS vs TIAN indistinguishable given thalamic/hypothalamic tumour invasion; treated with anakinra, siltuximab, IV **and** ICV corticosteroids, plus oral dasatinib to dampen CAR T-cell activity; resolved within 4 days.
- **No on-target off-tumour toxicity**: no painful neuropathy in any patient; autopsy in patient 1 showed lymphocytic infiltration and CAR transcript confined to tumour, not normal cortex, with much higher GD2 in tumour than normal brain.
- Serum LDH rose in all patients and tracked inflammation. Anakinra was detectable in CSF after IV dosing (i.e. IL-1 blockade reaches the compartment that matters).

Full arm A, 11 treated patients, 11 IV + 62 ICV infusions (Monje 2025, PMID 39537919) [GD2-clinical, ICV-clinical]

- **DLTs**: none at IV DL1; **three grade 4 CRS at IV DL2 ($3 \times 10^6/\text{kg}$)** → DL1 ($1 \times 10^6/\text{kg}$) is the maximally tolerated IV dose. Grade 4 CRS presentations: multi-pressor hypotension (1), pulmonary oedema with BiPAP/CPAP (1), intubation (1).

- **Zero DLTs across 62 ICV infusions.** CRS in 33.9% of ICV infusions (16 grade 1, 4 grade 2, 1 grade 3 — the grade 3 in the setting of urosepsis). **No ICANS after any ICV infusion**, despite higher CSF cytokines than after IV.
- TIAN in 91% after IV and 100% after first ICV infusion, 44/62 (71%) of ICV infusions overall; grade typically falls with successive infusions; **all TIAN reversed** under the management algorithm; no TIAN-attributed DLT. One sDMG patient had high-grade communicating hydrocephalus at peak inflammation.
- One **grade 5 intratumoural haemorrhage** in a region of known intratumoural vascular anomaly — contextualised against ~20% symptomatic intratumoural haemorrhage in DIPG by 12 months from diagnosis; causal attribution unresolved.
- One **grade 3 sensory neuropathy** (stocking-glove) after a first infusion, attributed in part to vitamin B6 toxicity and resolving after B6 cessation — the only peripheral-nerve event; **no antibody-type transient painful neuropathy in any patient.**
- Toxicity correlates: grade ≥ 2 CRS $\leftrightarrow$ higher plasma MCP-1 (CCL2) and IL-2; grade ≥ 2 TIAN $\leftrightarrow$ higher CSF MCP-1, IL-10, TNF.
- iCasp9 was never triggered (AP1903 not administered to any arm A patient).
- Trial design consequence: arms B and C now test **ICV-only** dosing with and without lymphodepletion — the sponsor's own read is that the systemic route is the toxic one.

GD2-expressing CNS tumours — GD2.CART $\pm$ constitutive IL-7 receptor (Baylor, NCT04099797)

(Bhat/Ahmed 2024, PMID 38771986) [GD2-clinical]

- 11 patients, IV, standard Cy/Flu lymphodepletion at all dose levels. **No DLTs.**
- GD2.CART alone (no C7R): no toxicity at all, and no durable benefit.
- C7R-GD2.CART: **TIAN grade 1 in 7/8 (88%)**, controlled with anakinra; CRS in 6/8 (75%), grade 1 in all but one, associated with rising IL-6 and IP-10.
- One patient with grade 2 CRS had heavy prior chemotherapy; subsequent DL2 patients received **split dosing** ($1.5 \times 10^7/\text{m}^2$ 5–7 days apart) and no further grade ≥ 2 CRS occurred — dose fractionation as a concrete mitigation.

Neuroblastoma — GD2-CART01 (third-generation CD28/OX40, iCasp9; Bambino Gesù)

Interim (Del Bufalo 2023, PMID 37018492) and final (Quintarelli/Locatelli 2025, PMID 40841488) [GD2-clinical]

- 27 $\rightarrow$ 54 children (35 on trial + 19 hospital exemption). Dose levels 3, 6, 10×10^6 CAR+/kg; **no DLTs**; MTD $10 \times 10^6/\text{kg}$.
- CRS in 20/27 (74%), **mild in 19/20 (95%)**.
- **Grade 3 ICANS in 4 children, "rapidly controlled" by rimiducid activation of iCasp9** — the only clinical dataset in which the GD2 suicide switch has actually been used therapeutically, at the cost of eliminating the product.
- No new safety signals with longer follow-up; persistence ≥ 12 months in 64%.

Neuroblastoma — other GD2 CAR-T products

- **4SCAR-GD2 (CD28/4-1BB/CD3ζ-iCasp9), 10 patients** (Yu 2022, PMID 34724115) [GD2-clinical]: no treatment-related deaths. Grade 3–4 events were **haematologic and attributed to Cy/Flu**, fully reversible without treatment in a median of 4 days (range 0–22). CRS grade 1–2 in 9/10, acute capillary leak in 4/10, **neuropathic pain grade 1–2** transient. Peak ferritin 3417 µg/L, IL-6 1000 pg/mL, IL-10 8014 pg/mL — i.e. biochemically real inflammation with clinically mild disease.
- **Second-generation GD2-CAR with escalating lymphodepletion, 12 patients** (Straathof 2020, PMID 33239386) [GD2-clinical]: grade 2–3 CRS in 2/6 at the highest dose after Flu/Cy; **no on-target off-tumour toxicity**; no objective responses at day +28.
- **ALLO_GD2-CART01, donor-derived, 5 children** (2025, PMID 39815015) [GD2-clinical]: **all five had grade 2–3 CRS**, one grade 2 neurotoxicity, and **moderate acute GVHD in 4/5** — allogeneic products add a toxicity class that autologous GD2 CAR-T does not have.
- **GD2-CAR.15 NKT cells (IL-15 co-expressing), 12 children** (Heczey 2020 PMID 33046868; 2023 PMID 37188782) [GD2-clinical]: **no DLTs**, MTD not reached; grade 3–4 haematologic events occurred before CAR-NKT infusion and were attributed to Cy/Flu; one grade 2 CRS resolved with tocilizumab.
- **PSMA + GD2 CAR-T for refractory glioma, 6 patients** (2025, PMID 40420236) [GD2-clinical]: CRS grade 1 in 3; **no central neurotoxicity**; haematologic toxicity in 5/6 (grade ≥2 thrombocytopenia 4/6, neutropenia 3/6), 3 infections associated with marrow suppression; explicitly "**no significant abnormalities in laboratory values, including liver enzymes, bilirubin levels, or electrolytes**", and none on long-term follow-up.
- **GD2⁺ medulloblastoma** (2024, PMID 38551501) is preclinical/translational with iCasp9 rescue demonstrated in mice (CAR⁺ cells 76.3% → 9.4% after AP1903) and murine NFL used as a neuroaxonal injury readout.

Pooled view

Systematic review/meta-analysis of 8 GD2 CAR-T neuroblastoma studies, 146 patients (2026, PMID 41530803) [GD2-clinical, pooled]:

- Most common any-grade AE: **anaemia 97.4%** (81.5–100). Most common grade ≥3: **neutropenia 93.5%** (72.7–100), then thrombocytopenia, anaemia, CRS.
- Efficacy: CR 39.6%, PR 15.8%.
- Interpretation caveat: these pooled cytopenia rates are dominated by **lymphodepleting chemotherapy**, not by GD2 target engagement, and the confidence intervals reach 100% because the studies are tiny. The correct reading is "expect near-universal transient marrow suppression whenever Cy/Flu is used", which is precisely the component ICV re-dosing omits.

Cross-trial signal for the ICV question

- Every GD2 CAR-T trial that gave cells **intravenously with lymphodepletion** reported CRS in most patients and grade 3–4 cytopenias.

- The only trial that gave cells **ICV without lymphodepletion** (Stanford arm A second and subsequent doses) reported no DLT in 62 infusions, no ICANS, mostly grade 1 CRS, and toxicity that was overwhelmingly intracranial and mechanical rather than systemic.
- No GD2 CAR-T trial has reported clinically significant transaminitis, hyperbilirubinaemia, AKI or electrolyte derangement as an attributed toxicity; where laboratory values were explicitly reported normal, the route was ICV or the CRS was low grade.

02 — Preclinical toxicity and on-target / off-tumour GD2 biology

The affinity trap

Richman 2018 (PMID 29180536) [preclinical] is the reference hazard for this antigen. Variants of the 14G2a scFv engineered for improved stability/affinity, including **E101K**, gave better antitumour activity against GD2⁺ neuroblastoma xenografts **and lethal CNS toxicity**: extensive CAR T-cell infiltration and proliferation in brain with neuronal destruction, an encephalitis localised to **cerebellum and basal brain regions that express only low amounts of GD2**.

Three conclusions that constrain every subsequent GD2 CAR design:

1. Toxicity was not driven by high-antigen tissue — it appeared where GD2 is low, i.e. affinity raised the CAR above the antigen-density threshold that normally protects normal tissue.
2. Antibody safety does not transfer to CAR safety at the same epitope: dinutuximab is approved, yet the same binder in a high-affinity CAR was fatal in mice.
3. The clinically used GD2 CARs deliberately keep **standard 14g2a affinity**, which is why the density-discrimination argument (Majzner 2022 autopsy: tumour GD2 ≫ normal brain GD2) is load bearing rather than rhetorical. Affinity-tuning literature in the collection (PMID 34966954) makes the same point generically.

Brainstem inflammation predicted the clinical problem

Preclinical H3K27M DMG models with GD2-CAR T cells (Mount 2018, PMID 29662203) [preclinical] showed clearance of tumour **and** brainstem oedema in a fraction of mice, progressing to obstructive hydrocephalus because of the neuroanatomy. This is the direct ancestor of the clinical TIAN management algorithm: the Stanford protocol excluded bulky thalamic and cerebellar tumours, mandated an Ommaya for ICP monitoring in DIPG, and pre-specified CSF removal, hypertonic saline, anti-cytokine agents and corticosteroids (Majzner 2022, PMID 35130560).

So the preclinical signal that mattered was **not** on-target off-tumour neurotoxicity — it was on-tumour inflammation in a space-limited compartment.

What normal-tissue GD2 does with antibodies

[GD2-antibody] evidence sets the expectation for target engagement in normal nerve:

- Anti-GD2 antibodies cause **dose-limiting on-target/off-tumour neuropathic pain** in most children (ACCELERATE Paediatric Strategy Forum, PMID 41240535; mechanism review PMID 34884452).
- Rat models reproduce anti-GD2-induced allodynia and it is reducible by pretreatment (DFMO, PMID 32697811); dinutuximab variants engineered for reduced neuropathy

induction exist (PMID 35792784); complement/Fc engineering and IgA formats are being used to dissociate efficacy from pain (PMID 37479484).

- FAERS disproportionality analysis of dinutuximab/dinutuximab β /naxitamab (PMID 38153627) confirms a broad real-world toxicity spectrum including signals absent from labels.

The clinical GD2 CAR-T experience does **not** reproduce this: no painful neuropathy in the Stanford DMG cohort (Monje 2025), none attributed in Straathof 2020, only transient grade 1–2 neuropathic pain in 4SCAR-GD2 (PMID 34724115). Plausible reasons — different effector mechanism (no complement activation on nerve), lower and non-repeated peak target occupancy on nerve, and antigen-density thresholding — but this is inference, not a demonstrated mechanism.

Preclinical work on making it safer

- **iCasp9/AP1903** eliminates activated CAR T cells preferentially and is built into GD2-CART01, 4SCAR-GD2 and the Stanford construct (PMID 25389405; medulloblastoma model PMID 38551501 showing CAR⁺ fraction 76.3% → 9.4% after AP1903).
- **Costimulation-only / logic-gated designs** to avoid activation on normal tissue (PMID 28341563, ALK AND-gate PMID 36396552).
- **Microenvironment-actuated and density-dependent ("velcro-like") CARs** claiming efficacy without toxicity (PMID 39841845, PMID 41005308); sonogenetic spatial control (PMID 40179881).
- Murine **neurofilament light chain** as a circulating neuroaxonal injury biomarker in CAR-T models (PMID 38551501) — the obvious translational readout for a "did we injure nerve?" question that clinical GD2 trials currently answer only by examination.

Transfer caveat

Xenograft models are lymphodepleted, immunodeficient and small; they cannot model human CRS (no functional myeloid compartment), and the mouse brainstem has different compliance and CSF dynamics. They have proven predictive for **spatial** risk (where inflammation is dangerous) and for **affinity/architecture** risk, and predictive for nothing about hepatic, renal or drug-interaction behaviour.

03 — The ICV route: what it removes, what it adds, and TIAN

What ICV administration removes

From the internal IV-vs-ICV comparison in NCT04196413 arm A (Monje 2025, PMID 39537919) [GD2-clinical, ICV-clinical]:

- **Dose-limiting CRS:** grade 4 CRS was the IV DLT at $3 \times 10^6/\text{kg}$; across 62 ICV infusions there was no DLT and CRS was absent in 41/62 infusions, grade 1 in 16.
- **ICANS entirely:** 0/62 ICV infusions, versus 5/11 patients after IV. This was explicitly unexpected because CSF cytokines are higher after ICV — consistent with ICANS being driven by systemic inflammation and endothelial/BBB injury from the blood side rather than by CSF cytokine concentration per se.
- **Lymphodepletion**, and with it the grade 3–4 cytopenias, febrile neutropenia and infection risk that dominate the AE tables of IV GD2 CAR-T trials. ICV re-dosing in arm A used no lymphodepletion.
- **Measurable systemic organ derangement:** no attributed hepatic, renal or electrolyte toxicity in the ICV experience; in the ICV-only B7-H3 trial the sole haematologic AE recorded during the DLT window was one grade 1 lymphocytopenia (Vitanza 2025, PMID 39775044).

What ICV administration adds or keeps

1. **TIAN** — near-universal on first ICV exposure in DIPG, then attenuating.
2. **Intracranial pressure and CSF-outflow risk** — the dangerous, minutes-to-hours toxicity.
3. **Hardware** — Ommaya/Rickham reservoir or CNS catheter as an eligibility requirement, with its own infection, malposition and revision burden; ICV trials require a catheter before enrolment (Vitanza 2025).
4. **Intratumoural haemorrhage** — the single DLT in ICV B7-H3 (grade 4, PMID 39775044) and the one grade 5 event in GD2 arm A (PMID 39537919). Both cohorts argue background DIPG haemorrhage rates (~20% by 12 months) make attribution ambiguous; neither can exclude a contribution.
5. **Repeat-dosing exposure** — ICV programmes give many infusions (arm A median 4, range 1–17; BrainChild-03 median 9, range 1–81 over up to 37.5 months), so cumulative and late toxicity, and anti-CAR immunity, become the relevant questions rather than single-infusion peak toxicity.

TIAN: definition, incidence, management

TIAN is **not** ICANS. It is neurotoxicity attributable to inflammation at the tumour site, graded by CTCAE, and split into two types (Majzner 2022, PMID 35130560; retrospective application in BrainChild-03, Ronsley 2026, PMID 41798119):

- **Type 1 — mechanical:** oedema, mass effect, obstruction of CSF flow → raised ICP, obstructive or communicating hydrocephalus, herniation risk. Treatment is **CSF removal via reservoir/EVD, hypertonic saline**, then anti-cytokine agents and corticosteroids.
- **Type 2 — electrophysiologic:** transient worsening of existing focal deficits without a mechanical cause (cranial-nerve dysfunction, trismus, hearing loss, ataxia, weakness), mapping to tumour neuroanatomy and often with corresponding T2/FLAIR change.

Incidence:

Cohort	Route	TIAN
GD2-CAR DMG arm A (n=11)	IV then ICV	91% of patients after IV; 100% after first ICV; 44/62 (71%) of ICV infusions; grade 3–4 in 8/62 ICV, 5/11 after IV; all reversible, no DLT
C7R-GD2.CART CNS tumours (n=8 C7R)	IV	grade 1 in 7/8 (88%), anakinra-controlled
B7-H3 CAR-T DIPG, BrainChild-03 (n=21)	ICV	16/21 (76%) met TIAN criteria; 49/152 infusions (32%); 34 grade 1, 14 grade 2, 1 grade 3; 1 needed CSF diversion; 13 had worsening pre-existing deficits

Practical points that recur across these cohorts:

- Symptoms are **predictable from tumour location** — pontine tumours give cranial-nerve and bulbar dysfunction; thalamic/cerebellar bulk was an exclusion criterion on the GD2 trial precisely because the preclinical models showed lethal hydrocephalus there.
- The therapeutic window is **short**: ICP 22 and 34 mmHg episodes in Majzner 2022 resolved "within minutes" of CSF drainage. This is why reservoir access, not pharmacology, is the primary intervention, and why ICP monitoring capability is an eligibility requirement rather than a nicety.
- **Fever after an ICV infusion is expected** and is not by itself CRS: BrainChild-03 explicitly interpreted fever as evidence of local immune activation (PMID 39775044).
- TIAN grade **falls with successive infusions** in both GD2 and B7-H3 cohorts — the first exposure is the risky one.
- Corticosteroids are used, but the algorithms put CSF diversion and IL-1/IL-6 blockade first, partly to protect CAR T-cell activity. Anakinra crosses into CSF after IV dosing (measurable CSF IL-1RA, PMID 35130560); ICV dexamethasone has been used in a refractory case (PMID 41479887 documents early intrathecal dexamethasone/methotrexate for ICANS), and Majzner 2022 used ICV steroids plus dasatinib as an off-label CAR "off-switch".

Comparator locoregional CAR-T trials (route-specific safety)

- **ICV B7-H3, DIPG, BrainChild-03 arm C** (PMID 39775044): 21 patients, **253 ICV doses**, up to 10×10^7 cells/dose declared the maximally tolerated dose regimen;

commonest AEs headache 81%, nausea/vomiting 81%, fatigue 62%, fever 57%, hiccups 29%; mostly grade 1-2; **hydrocephalus in 1 (grade 3)**; no ICANS; single DLT = grade 4 intratumoural haemorrhage. Multiyear repeat dosing was tolerable.

- **ICV B7-H3 for non-pontine DMG / other pediatric CNS tumours** (PMID 42503899) and **intracranial B7-H3 for recurrent GBM** (PMID 42562965) extend the same safety pattern.
- **Intraventricular/intrathecal bivalent EGFR + IL13R α 2 CAR-T in recurrent GBM** (PMID 38480922, PMID 40451950) and **CARv3-TEAM-E** — locoregional CNS CAR-T with early inflammatory neurotoxicity managed without dose-limiting systemic toxicity.
- **EGFR806-CAR locoregional in pediatric CNS tumours** (PMID 40070357) and **IL13R α 2 intraventricular** (BrainChild-01) complete the picture: across targets, the locoregional route consistently converts the toxicity problem from systemic-inflammatory to intracranial-mechanical.
- **Lymphodepletion in solid/brain-tumour CAR-T** (PMID 39542517) reviews whether the systemic conditioning that drives most measurable organ toxicity is even needed for locoregional delivery — the question arms B/C of NCT04196413 are now testing prospectively.

Route-specific unknowns

- Whether repeated ICV dosing causes cumulative ependymal/leptomeningeal injury, or CSF-flow impairment, is unmeasured; nobody reports longitudinal CSF dynamics.
- Device-related infection rates specific to CAR-T ICV programmes are not reported separately from general reservoir experience; the collection does contain post-CAR-T CNS infection reports (e.g. carbapenem-resistant *Klebsiella* CNS infection, PMID 41511674).
- Whether ICV dosing can cause CRS at higher cell doses than tested ($>10 \times 10^7$ per dose) is unknown — the grade 3 CRS in arm A ICV occurred with concurrent urosepsis, and CSF-to-blood cell egress is documented.

04 — CRS, ICANS and IEC-HS/HLH

CRS mechanism relevant to GD2 products

CRS is myeloid-amplified, not purely T-cell-derived: monocyte/macrophage **IL-1 and IL-6** drive the syndrome and IL-1 blockade separates CRS from neurotoxicity in models (Norelli 2018, PMID 29808007; Giavridis 2018, PMID 29808005). This is the mechanistic justification for anakinra being used alongside or ahead of tocilizumab in the GD2 CNS trials, and for anakinra's usefulness in TIAN, since IL-1RA reaches the CSF (PMID 35130560).

Grading is ASTCT consensus (Lee 2019, PMID 30592986); pediatric management follows dedicated guidelines (PMID 30082906). Endothelial activation and BBB disruption underlie severe neurotoxicity (Gust 2017 and related records in the neurotoxicity category), which is why endothelial-activation indices (EASIX and variants) recur in the collection as predictors of coagulopathy and ICANS.

Incidence in GD2 products

Product / route	CRS	Severe CRS	Neurotoxicity
GD2-CAR DMG, IV DL1 (n=3)	3/3	none	ICANS 1 (grade 2); TIAN 3/3
GD2-CAR DMG, IV DL2 (n=8)	8/8	3 grade 4 (DLT)	ICANS 4/8 (1 grade 3); TIAN 7/8
GD2-CAR DMG, ICV (62 infusions)	21 (33.9%)	1 grade 3 (with urosepsis)	ICANS 0 ; TIAN 44 (71%)
GD2-CART01 neuroblastoma (n=27)	20 (74%)	1/20 non-mild	grade 3 ICANS in 4/54 overall, reversed by rimiducid/iCasp9
C7R-GD2.CART CNS (n=8)	6 (75%)	1 grade 2	TIAN grade 1 in 7/8
4SCAR-GD2 neuroblastoma (n=10)	9 (90%)	none >grade 2	grade 1-2 neuropathic pain
GD2-CAR.15 NKT (n=12)	1 (grade 2)	none	none
ALLO_GD2-CART01 (n=5)	5	grade 2-3 in all	1 grade 2

Reading: CRS is essentially universal with IV administration of an active GD2 product, but severity is dose-driven and mostly low grade; the two high-severity signals are **IV dose escalation** (grade 4 CRS at 3×10^6 /kg in DMG) and **allogeneic products** (grade 2-3 CRS in 5/5 plus GVHD).

Anti-cytokine therapy actually used

- **Tocilizumab** (IL-6R) — first line for CRS in every GD2 trial.
- **Anakinra** (IL-1R) — used early and pre-emptively in the CNS trials, including for TIAN; the only agent with documented CSF penetration in this setting.
- **Siltuximab** (IL-6) — used in the refractory grade 4 ICANS case (PMID 35130560).

- **Corticosteroids** — IV and, in one case, ICV.
- **Dasatinib** — off-label pharmacological "pause switch" on CAR T-cell signalling (PMID 35130560); itacitinib (JAK1) has been evaluated for CRS cytokine reduction in GD2 CAR-T contexts (PMID 32998963).
- **iCasp9 + rimiducid/AP1903** — definitive but destroys the product; used clinically in 4 children with grade 3 ICANS on GD2-CART01 (PMID 40841488).

IEC-HS / carHLH

[CART-general]. Immune effector cell-associated HLH-like syndrome (IEC-HS, formerly carHLH) is the hyperinflammatory extreme: overall incidence after CAR-T ~3.5%, with ferritin elevation as the most reliable feature and **grade 3-4 ALT/AST elevation as part of the diagnostic picture** (PMID 41345813 review; ASTCT/consensus description PMID 36906275; carHLH after CD19 PMID 34263927; CD22 HLH-like CRS variant PMID 34525183; management PMID 41779733, PMID 41311365).

Why it matters for GD2 CAR-T:

- GD2 CAR-T trials report high ferritin (peak 3417 µg/L in 4SCAR-GD2) without formal HLH, so the syndrome is on the differential whenever fever persists with rising ferritin, falling fibrinogen and transaminitis.
- IEC-HS is **the** CAR-T scenario in which hepatic dysfunction becomes clinically meaningful rather than a laboratory finding, and where etoposide, ruxolitinib, anakinra or emapalumab (IFN-γ blockade, PMID 35917457, PMID 42136039) enter the picture — each with its own interaction and myelosuppression profile.
- H-Score and HLH-2004 criteria perform poorly in this population (PMID 35045791); use the IEC-HS framework instead.

No case of IEC-HS has been reported after ICV GD2 CAR-T in the published cohorts.

ICANS vs TIAN: the practical discrimination

- **ICANS**: encephalopathy, aphasia, tremor, seizures, decreased level of consciousness; diffuse EEG slowing; symptoms **not** confined to tumour neuroanatomy; usually follows systemic CRS by days; responds to steroids ± anakinra.
- **TIAN**: deficits map to tumour; may occur without systemic CRS; type 1 needs CSF drainage/osmotic therapy urgently.
- They are **not always separable**: in the patient with extensive thalamic/hypothalamic/insular tumour invasion, grade 4 ICANS and TIAN could not be distinguished (PMID 35130560). Where tumour is widespread, treat for both.

05 — Transient organ-function change: hepatic, renal/electrolyte, haematologic, coagulation, cardiopulmonary

This note answers the "what changes, by how much, for how long, and does it matter for other drugs?" question. Almost all quantitative data are [CART-general] (CD19/BCMA/CD22, IV route, lymphodepleted adults); GD2-specific data are sparse and reassuring. The bridge between them is **systemic inflammation grade**, not the target antigen: if CRS is grade ≥ 3 the organ changes below should be expected regardless of what the CAR binds.

Hepatic

Best single source: **Hepatic effects and potential DILI of CAR-T therapy** (2026, PMID 41345813), a systematic review of liver safety across CAR-T trials [CART-general].

- Grade ≥ 3 ALT elevation ($5\text{--}20\times$ ULN) ranged **0–8%** across 19 studies, with **11 of 19 reporting no grade ≥ 3 elevations at all**; AST behaves similarly.
- Reported AE incidence badly understates laboratory reality: in ZUMA-5 ($n=148$) increased ALT was reported as an AE in **8%** while the actual laboratory abnormality frequency was **58%**.
- **Hepatic dysfunction is the most common organ dysfunction in grade 3–4 CRS: 64% of patients** (vs cardiac dysfunction 36%). Another cohort reported "liver toxicity" in 31%.
- Grade 3–4 ALT/AST elevation occurs in **31–64%** of patients with severe CRS. **AST > ALT** is typical, suggesting an ischaemic/hypoperfusion contribution rather than direct hepatocellular toxicity.
- Bilirubin rise is uncommon and usually mild; in one pooled analysis of 137 patients, 12 (11%) had grade ≥ 3 total bilirubin and 11 of those 12 (92%) had CRS — i.e. bilirubin elevation is a marker of CRS severity, not of an independent hepatic syndrome.
- In IEC-HS, grade 3–4 transaminase elevation is very common and is part of the diagnostic construct; bilirubin $>ULN$ in $>50\%$ and $>2\times$ ULN in $\sim 33\%$.
- Rarer, genuinely hepatic entities do occur: acute liver failure without CRS/IEC-HS within 2 months, sinusoidal obstruction syndrome, early immune-related hepatitis, isolated grade ≥ 3 transaminitis, and late (beyond the acute window) transaminase elevation.

Pattern: transaminase elevation is common, tracks CRS grade, peaks with the inflammatory peak (days $\sim 5\text{--}14$ after IV infusion), is usually asymptomatic and resolves as CRS resolves. Bilirubin and synthetic function are usually preserved outside IEC-HS.

GD2-specific: no GD2 CAR-T trial reports hepatic toxicity as an attributed AE. The PSMA+GD2 glioma trial explicitly recorded **no abnormalities of liver enzymes or bilirubin**, acutely or on long-term follow-up (PMID 40420236). Anti-GD2 antibody regimens do include hepatic events (e.g. in dinutuximab-containing induction, PMID 37760578, and ^{131}I -MIBG + dinutuximab, PMID 40549985), but those are chemotherapy/radiopharmaceutical-containing combinations.

Renal and electrolyte

- Pooled incidence of AKI after CAR-T: **18.6%** across 22 cohort studies / 3,376 patients (17% in adults), with CRS incidence 75.4% in the same set (Kanduri systematic review, cited in PMID 39506012) [CART-general].
- **Transience is the rule:** in a 24-patient AKI series, onset was day +1 to +28 (13 of 22 post-infusion cases at day +7), and **renal function recovered in 19/24 (79%) within the first month** (PMID 38500492). Comparable series: AKI in 30% of 46 NHL patients (Gutgarts), 22% of 67 DLBCL patients, both "mild with fast recovery"; no excess versus auto-HCT in the largest comparison (232 CAR-T vs 414 auto-HCT).
- Risk factors: male sex, pre-existing CKD, **CRS**, and **ICANS grade ≥ 3** — i.e. AKI is a hypoperfusion/inflammation phenomenon, aggravated by nephrotoxic co-medication and contrast.
- AKI after CAR-T is associated with inferior survival in some series (PMID 40589093) — it is a severity marker even when it resolves.
- Mechanisms and management are reviewed from the nephrology side (PMID 39781479, PMID 40107382, PMID 41536632, ADQI consensus PMID 41361704), including CAR-T in dialysis-dependent and transplant recipients (PMID 41342049, PMID 34401733).
- Electrolyte disorders accompany CRS and immunotherapy generally (PMID 35063968): hyponatraemia (fluid resuscitation, SIADH, and — specific to this population — **hypertonic saline used for TIAN deliberately moves sodium**), hypophosphataemia, hypokalaemia, and tumour lysis where disease burden is high. FGF23-driven hypophosphataemia is described in HLH (PMID 42379468).
- **GD2/ICV-specific:** no attributed renal or electrolyte toxicity in any GD2 CAR-T trial; the ICV B7-H3 AE table contains no renal or electrolyte events at all (PMID 39775044). But **sodium management is an active intervention in TIAN**, so sodium, osmolality and fluid balance are on the monitoring list for reasons unrelated to nephrotoxicity.

Haematologic and coagulation

- In GD2 CAR-T trials the grade 3–4 haematologic events are attributed to **Cy/Flu lymphodepletion**, not to the CAR: fully reversible in a median 4 days (range 0–22) in 4SCAR-GD2 (PMID 34724115); in CAR-NKT the grade 3–4 cytopenias occurred before cell infusion (PMID 33046868).
- Pooled GD2 neuroblastoma data: any-grade anaemia 97%, grade ≥ 3 neutropenia 93% (PMID 41530803) — dominated by conditioning, tiny studies, very wide CIs.
- [CART-general] the durable problem is **ICAHT** (immune effector cell-associated haematotoxicity): biphasic and sometimes prolonged cytopenias with clonal-haematopoiesis and marrow-infiltration correlates (mechanism/management PMID 39973361; EHA/EBMT grading applied PMID 38181508; T-ICAHT for thrombocytopenia PMID 40258181; "How I treat cytopenias after CAR-T" PMID 36800563). **CAR-HEMATOTOX** predicts it (PMID 34166502; population validation PMID 40668622; comparative validation PMID 41229978), and ICAHT predicts infection and mortality.
- Coagulopathy: CRS-related coagulopathy is common with low absolute bleeding/thrombosis rates (PMID 41228344, PMID 34595461); DIC occurs and is under-recognised (PMID 36794498, national cohort PMID 42633171); endothelial activation predicts DIC/CRS/ICANS (PMID 36503182); bleeding and thrombosis meta-analysis

PMID 38574863. Practically: fibrinogen, D-dimer, platelets and INR move during CRS, which matters if the patient is anticoagulated or needs a neurosurgical procedure (reservoir placement, EVD, biopsy) during the inflammatory window.

- **ICV consequence:** without lymphodepletion, ICV cohorts show essentially no cytopenia (one grade 1 lymphocytopenia in 21 patients / 253 doses, PMID 39775044). That removes the transfusion, neutropenic-fever, antimicrobial-prophylaxis and coagulation-management burden almost entirely — and it removes the main constraint on giving concurrent myelosuppressive therapy.

Cardiopulmonary

The severe events in GD2 CAR-T are CRS-mediated: multi-pressor hypotension, pulmonary oedema requiring BiPAP/CPAP, and intubation — all three grade 4 CRS presentations at IV DL2 in the DMG trial (PMID 39537919). Capillary leak occurred in 4/10 in 4SCAR-GD2. [CART-general] cardiac dysfunction accompanies 36% of grade 3–4 CRS. No cardiopulmonary events were reported after ICV dosing beyond fever and one hypotension/hypertension pair in the B7-H3 AE table.

Infection

Lymphodepletion + steroids + anti-cytokine agents + prolonged cytopenias = infection risk, and infection confounds every toxicity attribution (the single grade 3 CRS after ICV GD2 dosing occurred with urosepsis; a grade 3 CRS-equivalent in an sDMG patient likewise). Infectious complications across novel CAR targets including GD2 are catalogued in PMID 34619768; fungal disease in the setting of GD2 CAR-T plus immunosuppression in PMID 33509115; CNS infection after CAR-T in PMID 41511674.

Summary table — transient change, magnitude, timing

Parameter	Direction/magnitude	Timing	Reversible?	Driver
ALT/AST	grade 1–2 common; grade ≥ 3 in 0–8% overall, 31–64% if CRS grade 3–4; lab abnormality rate $\gg$ reported AE rate	with CRS peak, ~d5–14 IV	Yes, with CRS	Inflammation/hypoperfusion
Bilirubin	usually normal; grade ≥ 3 ~11% in severe-CRS cohorts, 92% of those with CRS	CRS peak	Yes	CRS severity marker
Creatinine	AKI ~18.6% pooled, mostly KDIGO 1	day +1 to +28, mode day +7	79% recover ≤ 1 month	CRS, hypoperfusion, nephrotoxins
Sodium	hyponatraemia from fluids/SIADH; iatrogenic hyponatraemia from hypertonic saline for TIAN	during CRS/TIAN	Yes	Treatment as much as disease
ANC/platelets/Hb	grade 3–4 near-universal with Cy/Flu; prolonged in a minority (ICAHT)	days 0–28, may extend months	Mostly	Lymphodepletion $\gg$ CAR

Parameter	Direction/magnitude	Timing	Reversible?	Driver
Fibrinogen/D-dimer/ INR	CRS coagulopathy common, bleeding uncommon	CRS peak	Yes	Endothelial activation
LDH, ferritin, CRP	rise in all responders; ferritin to thousands	inflammation peak	Yes	Tumour lysis + inflammation
All of the above after ICV without lymphodepletion	essentially unchanged	—	—	Compartmentalised inflammation

06 — Drug interactions and pharmacokinetics: what transient inflammation does to concomitant medications

Framing. No GD2 CAR-T study has measured a drug-drug interaction. What exists is a well-quantified mechanistic literature showing that (a) acute inflammation suppresses cytochrome P450 activity, and (b) blocking IL-6 reverses that suppression. Both directions are relevant, because CAR-T patients receive an inflammatory insult and then, often within hours, an IL-6 blocker. Everything in this note is therefore **[mechanistic-PK]**: a strong basis for monitoring and for anticipating dose changes, not a measured GD2-specific interaction.

Direction 1 — CRS suppresses CYP-mediated clearance (drug levels rise)

- Human endotoxin/inflammation studies: antipyrine clearance **–35%** (95% CI 18–48) by day 2, hexobarbital **–20%**, theophylline **–20%** (reviewed in PMID 36059001).
- Surgery-induced acute inflammation: metabolic ratios fell by **CYP1A2 –53%, CYP2C19 –58%, CYP3A4 –61%** within 1–3 days postoperatively (same review).
- Midazolam clearance fell **65.4%** as CRP rose from 10 to 300 mg/L; CRP and IL-6 were covariates on midazolam CL.
- COVID-19 vs HIV: darunavir CL/F 4.1 vs 10.3 mL/h and AUC 161,387 vs 75,727 ng·h/mL, with **IL-6 the only significant covariate** (PMID 32856282).
- Bone-marrow-transplant engraftment: cyclosporine steady-state concentration rose **3.6-fold**, peaking concurrently with IL-6.
- PBPK, IL-6 10 → 100 pg/mL: simvastatin AUC **×2.36**, midazolam **×2.08**, omeprazole **×1.97**, dextromethorphan **×1.37**, warfarin **×1.29**, caffeine **×1.07** — CYP3A4 substrates most sensitive (PMID 34496043, PMID 38044486, PMID 40733104).
- **The closest analogue to CRS:** modelling of transient IL-6 elevation after **blinatumomab** gave maximal suppression of CYP3A4 **28% at 48 h lasting ~7 days**, CYP1A2 9%, CYP2C9 17%, with AUC ratios of 1.9 (simvastatin) and 1.7 (midazolam) — i.e. for a short cytokine spike, expect **<2-fold** exposure changes lasting about a week, magnitude driven by duration of cytokine elevation (PMID 38044486).
- Clinical corroboration in other settings: voriconazole exposure rises with inflammation and can produce phenoconversion of CYP2C19 genotype (PMID 39010708, PMID 41766069, PMID 42300127); pediatric-specific systematic review of inflammation on CYP activity (PMID 34462878).

Direction 2 — IL-6 blockade removes the suppression (drug levels fall)

- **Sarilumab 200 mg single dose + simvastatin** in RA: simvastatin Cmax **–46%**, AUC_∞ **–45%**; β-hydroxy-simvastatin acid Cmax **–35%**, AUC **–36%**; CRP fell >7-fold over the same week (PMID 27722854).
- Tocilizumab reduces simvastatin Cmax and AUC by the same mechanism (cited in the same paper); sirukumab decreased midazolam AUC 30–35%, omeprazole 37–45%, S-warfarin 18–19% while increasing caffeine AUC 20–34%.
- Continuous IL-6R antagonist therapy in RA changes CYP activity measurably (PMID 37831074); olokizumab shows the same class effect (PMID 38984761).
- Regulatory-style assessments exist for many pro-inflammatory-pathway biologics (PMID 36890677, PMID 27391296, PMID 38050329 for mosunetuzumab, PMID 38383116 for cytokine-CYP trends).

Net effect in a CAR-T patient: exposure to CYP3A4/2C19/1A2 substrates can rise during CRS and then fall abruptly when tocilizumab/siltuximab is given — a two-way, few-days-scale moving target rather than a steady shift.

Concomitant drugs where this plausibly matters in a GD2 CAR-T patient

Prioritised by likelihood of co-prescription in this population and narrowness of therapeutic index:

Drug class	Concern	Practical response
Antiseizure medications (levetiracetam mostly non-CYP; but phenytoin, carbamazepine, valproate, lacosamide)	Seizure prophylaxis is standard in CNS CAR-T; CYP substrates/inducers shift with inflammation, and steroids interact with several	Prefer non-CYP agents (levetiracetam); level-monitor if using phenytoin/carbamazepine/valproate
Azole antifungals (voriconazole, posaconazole, itraconazole)	Both victims (inflammation-driven overexposure, documented phenoconversion) and perpetrators (CYP3A4 inhibition)	Therapeutic drug monitoring during and after CRS; avoid assuming genotype-based dosing holds
Calcineurin/mTOR inhibitors (tacrolimus, sirolimus, ciclosporin)	3.6-fold ciclosporin rise documented with IL-6 peaks; used for GVHD in allogeneic GD2 products	Frequent trough monitoring, especially in ALLO_GD2-CART01-type settings with GVHD
Dexamethasone / methylprednisolone	CYP3A4 substrate and inducer; used at variable doses for T1AN/ICANS; also the drug most likely to blunt CAR efficacy	Expect variable exposure; use the minimum effective duration per the trial algorithms
Dasatinib (used as a CAR "off-switch")	CYP3A4 substrate — exposure will move with both CRS and tocilizumab	Not formally studied in this use
Opioids, midazolam, other sedatives	CYP3A4-dependent; children with brainstem disease already at risk of hypoventilation	Titrate to effect; anticipate reduced clearance during CRS
Anticoagulants (warfarin, DOACs)		

Drug class	Concern	Practical response
	CYP/coagulation double hit: warfarin S-enantiomer clearance changes plus CRS coagulopathy	INR/anti-Xa monitoring through the inflammatory window
Statins	Largest documented swing in either direction ($\times 2.36$ up with IL-6; -45% with sarilumab)	Consider holding during the acute window; the clinical consequence is usually small in this population
Chemotherapy given concurrently or as bridging	Hepatic/renal changes and cytopenias limit dosing; bridging intensity worsens haematopoietic recovery (PMID 40267180)	Sequence rather than overlap where possible
Nephrotoxins (aminoglycosides, amphotericin, contrast, NSAIDs)	AKI risk multiplies with CRS hypoperfusion	Avoid in the CRS window; dose to renal function daily, not on admission creatinine

Interactions specific to this therapy, not mediated by CYP

- Lymphodepletion dosing depends on renal function.** Fludarabine is renally eliminated; the first population-PK model built exclusively in CAR-T recipients found weight, **eGFR** and CAR construct type as significant covariates on clearance, with reduced clearance below eGFR 120 mL/min/1.73 m² (PMID 41471106). Fludarabine over-exposure is linked to treatment-related mortality in HSCT, and under-exposure to weaker lymphodepletion. So a transient pre-infusion creatinine rise is not a cosmetic finding — it changes the conditioning exposure. ICV re-dosing without lymphodepletion sidesteps this entirely.
- Corticosteroids and CAR T-cell activity.** Steroid exposure before or during infusion is an exclusion/deferral criterion in the GD2 DMG trial and CIS scoring was deferred until ≥ 7 days after steroid discontinuation (PMID 39537919). The interaction of interest is pharmacodynamic — steroids impair the therapy — and this is why the TIAN algorithm reaches for CSF drainage, hypertonic saline and IL-1/IL-6 blockade first.
- Hypertonic saline** for TIAN alters sodium and free-water handling, which changes the distribution/level of drugs sensitive to volume status and affects concurrent antiseizure agents.
- Anti-cytokine agents are themselves immunosuppressive** (anakinra, tocilizumab, siltuximab, emapalumab, ruxolitinib, etoposide for IEC-HS), stacking with lymphodepletion for infection risk and with cytopenias for marrow recovery.
- Therapeutic proteins do not have classic CYP interactions themselves** — the interaction is always via the cytokine milieu (PMID 36890677, PMID 27391296). Do not look for a "CAR T-cell DDI"; look for an inflammation-mediated one.

What is genuinely unknown

- Whether **ICV** administration, which raises CSF cytokines but not plasma cytokines, produces any systemic PK effect at all. Mechanistically it should be minimal; nobody has measured it.

- The magnitude and duration of CYP suppression during CAR-T CRS specifically — the blinatumomab transient-IL-6 model (~28% CYP3A4 suppression for a week) is the best available proxy but CAR-T CRS is longer and more intense than blinatumomab's cytokine spike.
- Whether inflammation-driven changes in **renal glomerular filtration** during CRS (modelled in PMID 37715056) are large enough to require dose adjustment of renally cleared co-medications beyond routine creatinine-based dosing.

07 — Monitoring and mitigation

Everything here is derived from what the published GD2 and locoregional CNS CAR-T protocols actually did, plus general CAR-T guidance (ASTCT grading PMID 30592986; pediatric management PMID 30082906; ICAHT grading PMID 38181508; IEC-HS management PMID 41779733). It is a synthesis of published practice, not a protocol.

Structural safeguards used in the GD2 CNS programme

- **Eligibility geography:** exclusion of bulky thalamic or cerebellar tumours (preclinical lethal hydrocephalus at those sites) and of clinically significant dysphagia as a marker of medullary dysfunction (PMID 39537919).
- **No corticosteroids at enrolment**, and CIS scoring deferred ≥ 7 days after steroid discontinuation.
- **Ommaya/Rickham reservoir placed before treatment** in DIPG, for ICP measurement and therapeutic CSF removal — routine plus symptom-prompted measurements. ICV trials of other targets require a CNS catheter as an eligibility criterion (PMID 39775044).
- **Pre-specified TIAN algorithm:** CSF removal $\rightarrow$ hypertonic saline $\rightarrow$ anti-cytokine agents $\rightarrow$ corticosteroids, with neurocritical-care availability.
- **iCasp9 suicide gene** in the construct, with rimiducid/AP1903 available (used in 4 children with grade 3 ICANS on GD2-CART01, PMID 40841488; never needed in DMG arm A).
- **Dose separation of route:** IV DL1 ($1 \times 10^6/\text{kg}$) as MTD, ICV flat doses $10\text{--}30 \times 10^6$ without lymphodepletion, $\geq 21\text{--}28$ days between infusions, requiring circulating CAR $< 5\%$ and toxicity resolved to $< \text{grade } 2$ before re-dosing.

Monitoring that the literature justifies

Neurologic / intracranial (both routes; the priority for ICV)

- Structured neurological examination at fixed timepoints, with a quantified improvement score so that worsening is detectable against a moving baseline.
- ICP measurement via reservoir — scheduled and on symptoms. Documented episodes: 22 mmHg and 34 mmHg with resolution within minutes of drainage.
- Low threshold for MRI (T2/FLAIR at the tumour site correlates with focal TIAN symptoms) and for EEG when encephalopathy is present (triphasic waves/diffuse slowing seen in the grade 4 ICANS case).
- Explicitly grade CRS, ICANS and TIAN **separately** at every infusion; do not collapse fever after ICV dosing into "CRS".

Systemic inflammation (IV route, and any febrile ICV patient)

- CRP, ferritin, fibrinogen, D-dimer, LDH, triglycerides; cytokine panel where available. Rising ferritin with falling fibrinogen and transaminitis $\rightarrow$ evaluate for IEC-HS.
- Blood counts; anticipate Cy/Flu nadir; use CAR-HEMATOTOX-type baseline risk assessment where the patient is heavily pretreated.

- Liver panel (ALT, AST, bilirubin, albumin, INR) through the CRS window — noting that grade 1–2 transaminase change is expected and does not by itself require intervention, while AST>ALT with hypotension suggests hypoperfusion, and bilirubin rise flags CRS severity or IEC-HS.
- Creatinine/eGFR **daily during the acute window**, with electrolytes including sodium, potassium, phosphate, magnesium; urine output and fluid balance. Sodium and osmolality are mandatory if hypertonic saline is used.
- Culture-directed infection workup for every fever, since infection both mimics and precipitates CRS (the two grade 3 CRS events in the ICV/sDMG experience were confounded by urosepsis).

Concomitant medications

- Reconcile the medication list against CYP3A4/2C19/1A2 dependence at baseline (see note 06) and again when tocilizumab/siltuximab is given.
- Therapeutic drug monitoring during and for ~1 week after the inflammatory peak for azoles, calcineurin/mTOR inhibitors, and any narrow-index CYP substrate; INR/anti-Xa if anticoagulated.
- Recalculate renally cleared drug doses on the current creatinine, not the admission value; this includes fludarabine exposure planning before conditioning (PMID 41471106).

Mitigation strategies with evidence behind them

Strategy	Evidence	Note
Use the ICV route for re-dosing, without lymphodepletion	62 ICV infusions, no DLT, no ICANS, CRS mostly grade 1, no organ-function signal (PMID 39537919); 253 ICV doses of B7-H3 with one DLT (PMID 39775044)	The single largest toxicity reduction available
Cap the IV dose	Grade 4 CRS at 3×10^6 /kg vs none at 1×10^6 /kg in DMG	MTD is route-specific
Split/fractionate dosing	No further grade ≥ 2 CRS after switching to 1.5×10^7 /m ² split 5–7 days apart (PMID 38771986)	Cheap, no engineering needed
Keep scFv affinity standard	Fatal encephalitis with high-affinity E101K in mice (PMID 29180536)	Antigen-density discrimination is the safety mechanism
Exclude high-risk tumour geography	Preclinical hydrocephalus in thalamic/cerebellar disease	Now standard eligibility
Pre-emptive IL-1 blockade (anakinra)	Used prophylactically/early for TIAN and CRS in GD2 CNS trials; CSF-penetrant	IL-1 mechanism separates CRS from neurotoxicity (PMID 29808007, PMID 29808005)
CSF drainage + hypertonic saline first for type 1 TIAN	Minutes-scale reversal of ICP events	Preserves CAR activity vs steroids
iCasp9 + rimiducid	Reversed grade 3 ICANS in 4/54 children (PMID 40841488)	Definitive but ablates the therapy
Pharmacological CAR pausing (dasatinib)	Used in refractory grade 4 ICANS (PMID 35130560); itacitinib explored (PMID 32998963)	Reversible alternative to iCasp9; not formally trialled

Strategy	Evidence	Note
Engineering: logic gates, affinity tuning, microenvironment-actuated CARs	Preclinical only (PMID 28341563, PMID 36396552, PMID 39841845, PMID 41005308)	Not yet in GD2 patients

Gaps worth closing

1. **No prospective organ-function dataset for ICV GD2 CAR-T.** The claim "ICV spares the liver and kidneys" rests on absence of reported AEs, not on published serial laboratory values.
2. **No measured DDI or CYP-probe study in any CAR-T CRS cohort,** let alone GD2 or ICV. A midazolam or 4-drug-cocktail probe study around CRS and tocilizumab administration would settle a question currently answered by PBPK extrapolation.
3. **No standardised TIAN grading validated prospectively.** BrainChild-03's retrospective application of TIAN criteria (PMID 41798119) showed most events could not be classified as type 1 vs type 2 — the distinction that determines whether CSF diversion is indicated.
4. **Repeat-dosing late toxicity** (ependymal injury, CSF-flow changes, anti-CAR immunity, device complications) is unreported despite cohorts receiving up to 81 doses over >3 years.
5. **No circulating neuro-injury biomarker** in clinical GD2 CAR-T; NFL is used in mouse models (PMID 38551501) and would directly address the residual on-target off-tumour question.
6. **Attribution of intratumoural haemorrhage** — two cohorts, one grade 5 and one grade 4 event, each argued to be background DIPG natural history. Only a larger denominator resolves this.

08 — The myeloid/microglial axis: recruitment, states, and its relation to TIAN

The clinical toxicity of GD2 CAR-T in the CNS correlates better with **myeloid** readouts than with T-cell readouts. This note collects what is actually measured, because it is the mechanistic bridge between "CAR T cells engage tumour" and "the patient's brainstem swells".

What is measured in GD2 CAR-T patients

Tumour tissue (autopsy, DIPG-1; PMID 35130560) [GD2-clinical]

- Prominent **microglial and other myeloid infiltration of tumour**: CD163⁺ myeloid cells throughout tumour tissue, and Iba1⁺ myeloid cells densely infiltrating H3K27M⁺ regions on confocal microscopy.
- In the **unaffected cortex** of the same patient, CD163 staining showed only microglia in their resting perivascular position — **no reactive microglia/macrophages**. Combined with CAR transcript and lymphocytic infiltrate confined to tumour, and tumour GD2 $\gg$ normal-brain GD2, this is the strongest single piece of human evidence that the inflammatory response is spatially restricted to tumour rather than provoked diffusely in normal brain.

CSF single-cell RNA-seq across routes (PMID 35130560) [GD2-clinical, ICV-clinical]

- CSF myeloid cells resolved into **seven clusters** — monocytes, microglia and macrophages with distinct functional signatures, plus a proliferating myeloid cluster (myeloid identity called on AIF1, CSF1R, CX3CR1, CD14, CD68, CD163).
- **Route-dependent myeloid states**:
 - after **ICV** administration, at peak inflammation, an **interferon-response myeloid population** dominates;
 - after **IV** administration and at late timepoints, myeloid cells instead express **phagocytosis and lipid-metabolism programmes** with a strongly **immune-suppressive** profile, aligning with **disease-associated microglia (DAM)**, MDSC and **axon-tract-associated microglia** signatures.
- The patient with the most severe intracranial events (DIPG-1) had **differentially increased interleukin, chemokine and neutrophil-degranulation pathways** in the CSF myeloid fraction relative to the other patients at peak inflammation.

So the ICV route does not merely relocate inflammation — it elicits a **different myeloid state** (IFN-responsive/activating) than the IV route (suppressive/DAM-like). Whether the ICV state is mechanistically responsible for the higher CSF cytokines with lower systemic toxicity is untested.

Cytokine correlates of toxicity (PMID 39537919) [GD2-clinical]

- Grade ≥ 2 **CRS** correlated with higher **plasma MCP-1/CCL2** — a monocyte-recruiting chemokine — plus a trend for IL-2.
- Grade ≥ 2 **TIAN** correlated with higher **CSF MCP-1/CCL2**, IL-10 and TNF- α .

- The authors' own reading: the association of severe TIAN with myeloid-response chemokines "raises the prospect that **myeloid cells may contribute to neurological symptoms** induced by CAR T cell therapy for CNS tumours, a hypothesis that requires testing in future studies."

That is the current state of the evidence: **correlative and explicitly hypothesis-generating**. No GD2 CAR-T study has depleted or blocked myeloid recruitment and measured the effect on TIAN.

Why a myeloid mechanism is plausible

[CART-general] mechanism, transferable because it is antigen-agnostic:

- CRS itself is **macrophage-mediated**: monocyte/macrophage IL-1 and IL-6 drive the syndrome, and **IL-1 blockade abates CRS while separating it from neurotoxicity** (Giavridis 2018, PMID 29808005; Norelli 2018, PMID 29808007). Monocyte depletion has been explored as a CRS intervention (PMID 34570442 review; monocyte-depletion preprint, DOI 10.1101/2020.11.22.20232801).
- In ICANS, **CD14⁺ myeloid cells are found in CSF**, and the proposed cascade is endothelial activation → BBB disruption → immunocyte infiltration → astrocyte injury and **microglial activation** → neuronal dysfunction (PMID 35871757). Autopsy in CAR-T cerebral oedema showed microglial activation, extensive in one case and perivascular in another (cited in PMID 35871757).
- Cytokine-level detail (PMID 33391257): **GM-CSF** secreted by CNS-infiltrating helper T cells activates microglia, and GM-CSF neutralisation reduced brain contrast enhancement in a CAR-T xenograft; **IFN- γ** drives microglial activation and antigen presentation; **IL-6** induces microglial proliferation; **IL-15** induces microglial cytokine production, and increased astrocytic IL-15 worsened cerebral oedema in mice — relevant because IL-15-armoured products exist in the GD2 space (GD2-CAR.15 NKT). Microglia also retain **epigenetic memory** of inflammatory stimuli, a proposed substrate for prolonged neurologic sequelae.
- Preclinically in GD2/DMG models, tumour clearance produced **brainstem oedema and obstructive hydrocephalus** (PMID 29662203) — the mass-effect endpoint that TIAN type 1 describes.

Is microglia a target or an amplifier?

An important distinction for the off-target question: nothing in this corpus shows GD2 CAR T cells **targeting** microglia. GD2 in normal CNS is described on neurons, astrocytes and peripheral nerve fibres, not as a microglial marker, and the only demonstrated GD2 CAR-mediated normal-CNS injury is the high-affinity **E101K** encephalitis, where CAR T cells infiltrated and proliferated in low-GD2 cerebellar and basal brain regions with neuronal destruction (PMID 29180536) — a T-cell-driven lesion, with myeloid involvement not separately characterised in that report.

Practical framing:

- **Amplifier (evidenced)**: myeloid recruitment and microglial activation downstream of CAR-T engagement, correlating with CRS (plasma CCL2) and TIAN (CSF CCL2/TNF/IL-10).
- **Target (no evidence)**: GD2 CAR-T killing microglia.
- **Consequence**: interventions aimed at the myeloid arm — anakinra (IL-1R), tocilizumab/siltuximab (IL-6), and in the hyperinflammatory extreme emapalumab (IFN- γ) or ruxolitinib — are the mechanistically-matched drugs for TIAN and CRS, whereas corticosteroids act broadly and also blunt the CAR. This is exactly the order the GD2 CNS protocols use.

Open questions this would answer if measured

1. Does blocking **CCL2/CCR2** or GM-CSF (lenzilumab-type) reduce TIAN without reducing antitumour effect? Untested in GD2 CAR-T.
2. Do the **DAM/MDSC-like suppressive myeloid states** seen after IV dosing and at late timepoints act as a resistance mechanism as well as a toxicity mechanism?
3. Does the attenuation of TIAN over successive ICV infusions reflect **myeloid-state adaptation** (tolerisation) or simply less tumour left to inflame?
4. Is there measurable **microglial injury** (e.g. sTREM2, GFAP, NfL in CSF) after ICV GD2 CAR-T? No GD2 trial reports neuroglial-injury biomarkers, although CSF/serum **GFAP and NfL** track neurotoxicity and endothelial dysfunction after CD19 CAR-T (PMID 41553539) and murine NfL has been used preclinically (PMID 38551501).

09 — Resistance mechanisms and failure modes

Toxicity is one axis of failure; this note collects the other one. Evidence labels as elsewhere ([GD2-clinical], [preclinical], [CART-general], [GD2-antibody]).

0. How well does it actually work?

- **Neuroblastoma, meta-analysis of 8 trials / 146 patients** (PMID 41530803) [GD2-clinical]: pooled CR **39.6%** (21.2–58.0), PR 15.8%, SD 30.8%, PD 20.9%. Authors' own verdict: efficacy "moderate"/limited.
- **GD2-CART01, final phase 1/2 + hospital-exemption cohort, 54 children** (PMID 40841488; interim PMID 37018492) [GD2-clinical]: ORR **66%**, CR 37–40% across timepoints, 5-year OS **42.7%** in the trial cohort; in the low-disease-burden target population ORR **77%**, 5-year OS 68%, EFS 53%. Persistence ≥ 12 months in **64%** (interim median persistence only 3 months, range 1–30).
- **CNS disease is worse than neuroblastoma**: DMG responses are transient tumour-volume reductions and neurological improvements, not remissions (PMIDs 35130560, 39537919, 38771986).
- **CAR-NKT** (PMID 37188782): ORR 25% (3/12).

1. Failure before the cells are ever infused (attrition)

Under-discussed but quantitatively the largest failure mode in DMG [GD2-clinical]:

- Stanford NCT04196413 (PMID 39537919): of the screened population, **36 patients progressed to ineligibility while on the waitlist or chose other options**; of 13 enrolled, **2 were removed before treatment** for rapid progression and performance-status decline. In the first report (PMID 35130560), a spinal DMG patient progressed during manufacturing and had to be treated on a compassionate eIND.
- Manufacturing itself is **not** the bottleneck: GD2-CART01 reported no manufacturing failures in 27 patients, and Stanford manufactured successfully for all patients.
- **T-cell fitness at collection is**: GD2-CART01 outcomes were substantially better when apheresis was performed **at diagnosis** rather than late, and after **1–2 prior lines** versus ≥ 3 (PMID 40841488). Prior cytotoxic therapy is therefore a resistance mechanism acting on the drug product, not the tumour.

2. Loss of function: exhaustion and short functional persistence

- The GD2 CAR is a **structurally exhaustion-prone construct**: the 14g2a scFv drives ligand-independent tonic signalling and early exhaustion, mitigated by 4-1BB costimulation (Long 2015, cited in PMID 35130560) and reversible by **transient rest**, including dasatinib-induced rest with epigenetic remodelling (Weber 2021, same citation set) [preclinical]. This is the mechanistic reason GD2 CAR design keeps changing.
- In the Stanford product, CAR⁺ cells already expressed significantly more **PD-1** than CAR[−] cells at release (CD39, LAG3, TIM3 not different) [GD2-clinical].

- **C7R trial** (PMID 38771986) is the cleanest in-human demonstration of persistence-limited failure: plain GD2.CART patients had neurologic improvement lasting **≤3 weeks**, progressed, and **none qualified for a second cycle**; adding a constitutively active IL-7 receptor gave longer PFS ($P < .005$), 7/8 eligible for repeat cycles, PR in 2/7 DMG — **but most still progressed eventually**.
- **CAR-NKT** (PMID 37188782): response tracked product composition (**CD62L⁺** frequency higher in responders), and peripheral CAR-NKTs upregulated **BTG1**, a driver of exhaustion-associated hyporesponsiveness; BTG1 knockdown restored control of metastatic disease in mice.
- Response correlates in DMG were weak and product-independent: no CD4:CD8 difference between the 8 responders and 3 non-responders; responders had higher post-IV plasma **IL-2** (PMID 39537919).

3. Compartmental failure: the drug does not reach all the disease

The single most instructive case in the corpus (spinal DMG-1, PMID 35130560) [ICV-clinical]:

- After ICV dosing her **spinal cord tumour regressed** with motor/urinary improvement and resolution of neuropathic pain, while a **temporal lobe metastasis did not respond at all**;
- the resected non-responding lesion showed **high and uniform GD2 by flow cytometry**;
- both compartments progressed by day +75.

So failure there was **access/delivery**, not antigen. ICV distributes through CSF and reaches CSF-adjacent and leptomeningeal disease better than bulky parenchymal disease remote from the ventricles — which is also why route optimisation (IV priming then ICV, multi-route, intratumoural) is the open question the trialists themselves name. Related, from the same corpus [CART-general]: poor trafficking into solid tumours is one of the three canonical explanations for the liquid-versus-solid efficacy gap (PMID 39537919 discussion).

4. Antigen-side resistance

- **No confirmed GD2-negative escape** has been documented in these GD2 CAR-T cohorts — unlike CD19, where antigen loss is the dominant relapse mechanism. The recurrent pattern is instead **antigen retained, response lost**. This is a weak negative: re-biopsy and autopsy antigen profiling was performed in only a handful of patients, so escape is unexcluded rather than excluded.
- **Antigen density thresholds** matter mechanistically: CAR efficacy is governed by target and receptor density (Walker 2017, cited in PMID 35130560), so partial GD2 downregulation below the activation threshold could produce escape without producing an antigen-negative tumour [preclinical].
- Density is also the axis being engineered deliberately — velcro-like density-dependent targeting of tumour-associated carbohydrate antigens (PMID 41005308) — which trades sensitivity for safety and therefore risks converting a safety margin into a resistance mechanism.
- [GD2-antibody] For dinutuximab, ~40% of patients fail or become resistant; tumour-derived **small extracellular vesicles** were shown to induce that resistance, reversible

with the farnesyltransferase inhibitor **tipifarnib** (PMID 35483745). Whether sEVs blunt CAR-T as well as ADCC is untested.

5. Microenvironmental suppression

- **MDSC/G-CSF axis** (PMID 39695851) [preclinical, GD2-specific]: CAR.GD2 T cells cleared disseminated rhabdomyosarcoma and orthotopic osteosarcoma, but co-presence of MDSCs **abolished long-term control**; tumour-derived **G-CSF** was identified as a key driver of MDSC expansion — a directly druggable node.
- **Myeloid suppression in neuroblastoma** more broadly: M-MDSC and PMN-MDSC are expanded in murine and human disease (PMID 41972140); **IGF2BP1** maintains the immunologically cold, immunosuppressive TME and its knockdown converts tumours to immunogenic (PMID 41328513) [preclinical].
- **In-patient evidence of a suppressive myeloid state**: CSF single-cell RNA-seq after **IV** dosing and at late timepoints showed phagocytic/lipid-metabolic myeloid cells with **DAM- and MDSC-like immunosuppressive signatures**, whereas peak ICV timepoints were interferon-responsive (PMID 35130560) — i.e. the same myeloid axis implicated in TIAN (see [notes/08](#)) plausibly also terminates the response. Untested causally.
- Standard countermeasures under study in this space: TGF- β dominant-negative receptors and armoured CARs (PMID 34572932 review; TGF- β -insensitive armouring validated clinically for PSMA, PMID 35314843), cytokine armouring (C7R, IL-15), microenvironment-actuated CARs (PMID 39841845), oncolytic-virus priming for DMG (PMID 42100144).

6. Iatrogenic and design-imposed failure (toxicity management defeating efficacy)

This is where [notes/07](#) and this note collide:

- **Corticosteroids** given for TIAN/ICANS blunt CAR-T activity; **dasatinib** deliberately switches CAR signalling off; **rimiducid/iCasp9** ablates the product outright — it rapidly resolved grade 3 ICANS in four GD2-CART01 children (PMID 40841488), at the cost of the therapy.
- **Toxicity capped the delivered dose**: grade 4 CRS at IV DL2 made **DL1 the MTD**, so the IV arm cannot be dosed at the level tested for efficacy (PMID 39537919).
- **ICV without lymphodepletion** removes systemic toxicity but also removes the homeostatic-cytokine niche that supports expansion; the trialists explicitly list "the role of lymphodepleting chemotherapy" as unresolved.
- **TIAN limits re-dosing cadence** even when re-dosing is the strategy that works — although TIAN severity attenuated with successive ICV infusions.
- Repeated ICV delivery depends on **hardware** (reservoir/EVD), so device complications are a route to losing the therapy rather than merely an adverse event.

7. What would resolve the ambiguities

1. Paired pre/post antigen profiling (GD2 density by flow, not just IHC positivity) at progression — the only way to separate density-threshold escape from functional failure.
2. CSF CAR-T phenotyping at progression versus peak, to test whether local exhaustion or myeloid suppression dominates.
3. A trial arm testing **MDSC/G-CSF or CCL2 blockade** alongside GD2 CAR-T; the preclinical rationale (PMID 39695851) is unusually direct and untested in humans.
4. Route comparison in patients with both CSF-adjacent and parenchymal lesions — the spinal DMG-1 pattern predicts a dissociation that is measurable.
5. Reporting apheresis timing and prior-line count as prespecified covariates; GD2-CART01 suggests they dominate outcome.

Appendix: full reference list

All 852 curated records, grouped by primary toxicity domain and ordered by year (descending). Records without a PMID are conference abstracts.

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